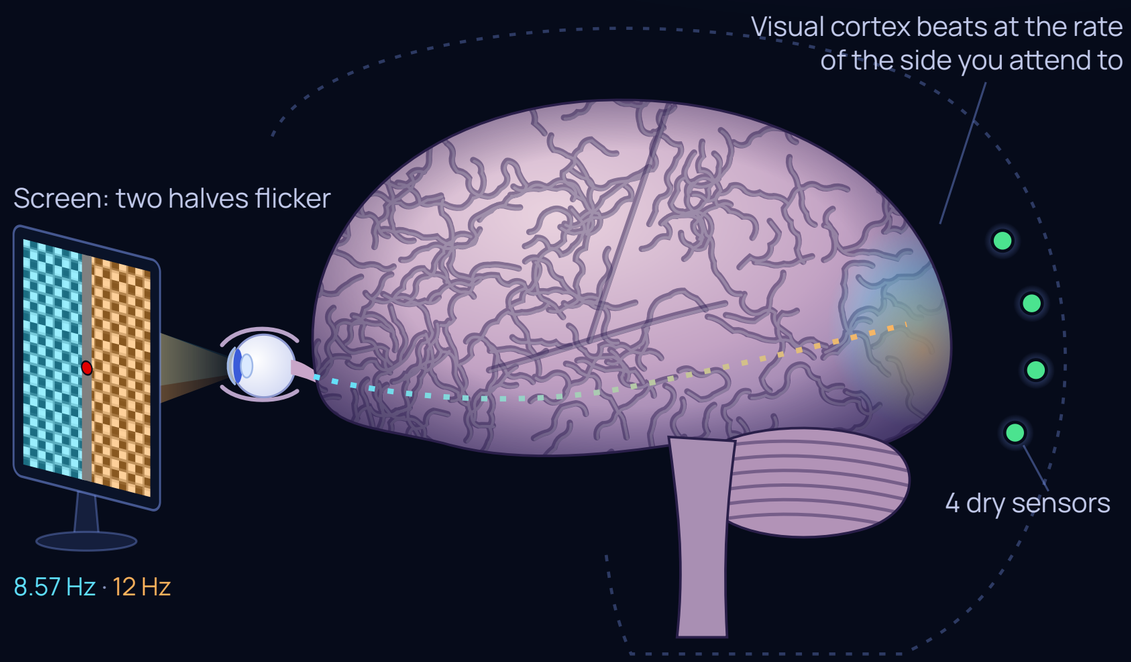
USF BCI CLUB BUILD SPRINT · BIOTECH TRACK

StillHere

Is anyone in there?

Checking if someone who can't move can still follow an instruction, using a laptop screen and an EEG headset.

`g.tec Unicorn BCI Core-4` `4 dry sensors` `a laptop screen`

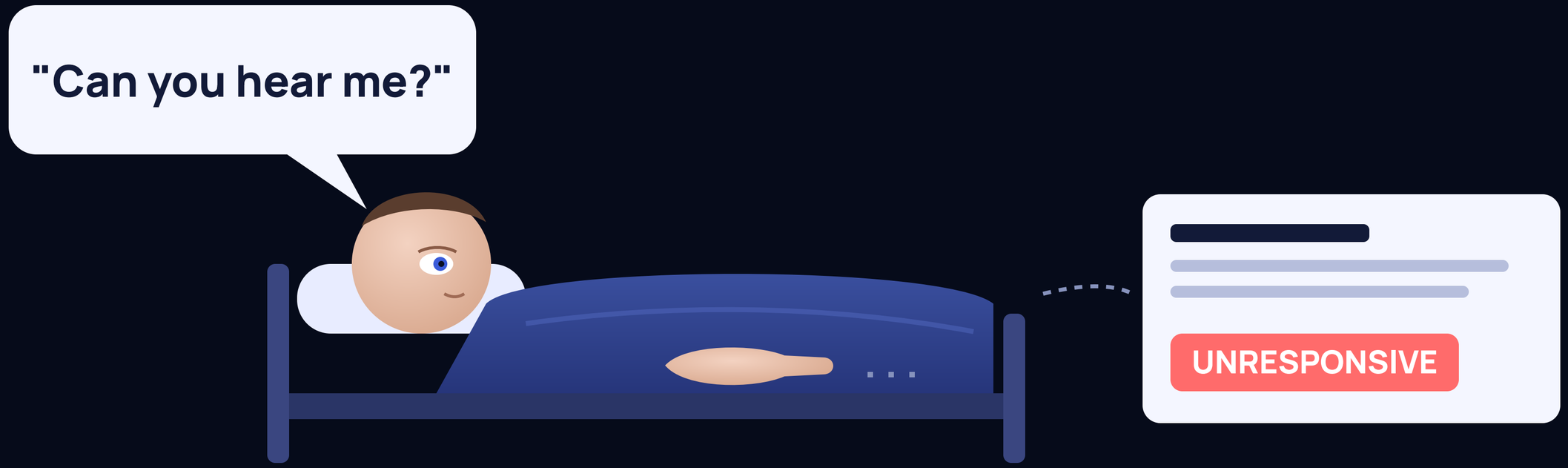


THE PROBLEM

Aware inside, unable to show it.

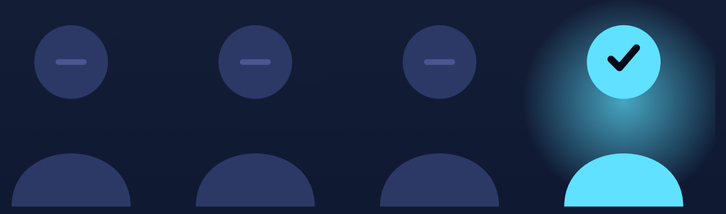
After a stroke, a brain injury or a cardiac arrest, some patients have their eyes open but don’t respond. Bedside tests all need some kind of movement.

If someone is **aware but can’t move**, they look the same as someone who isn’t.



BACKGROUND

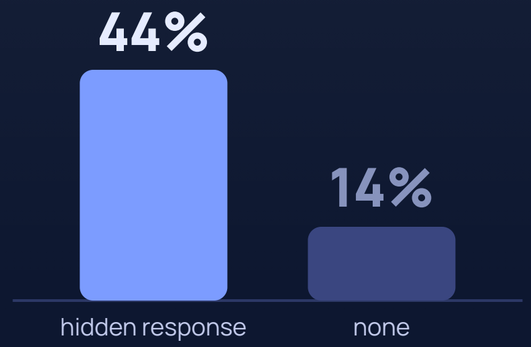
How common is this?



1 in 4

patients who **didn't respond** to commands could still do imagined-movement tasks on fMRI or EEG.

NEJM 2024 · 241 patients



of ICU patients with hidden brain responses were **living independently a year later** (vs 14%).

NEJM 2019 · 104 patients

~40%

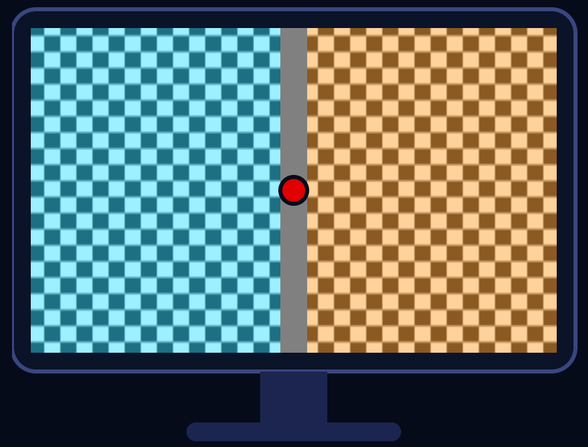
of patients diagnosed as vegetative were actually **minimally conscious**.

Brain Sciences 2020 review

THE IDEA

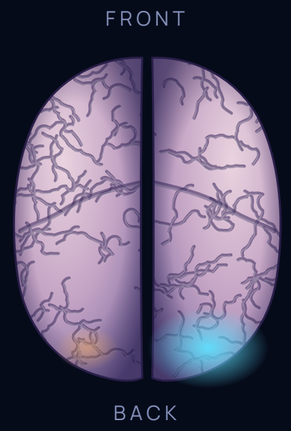
The brain signal we use (SSVEP)

When you look at a blinking light, the back of your brain picks up the same rhythm. If you pay attention to one side without moving your eyes, that side's rhythm gets a bit stronger.



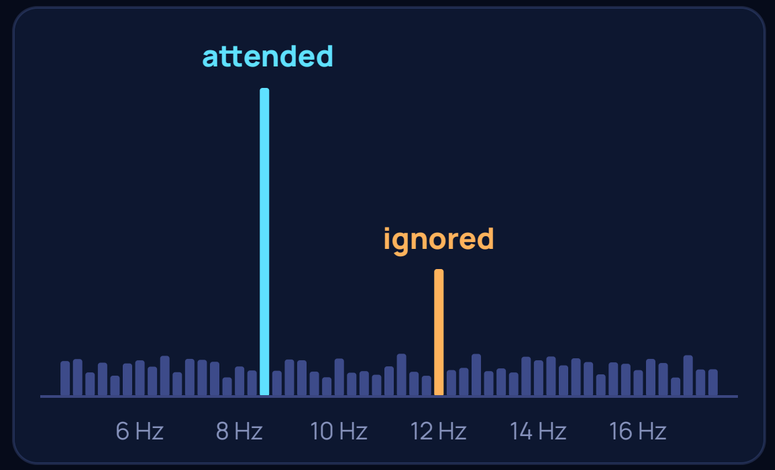
8.57 Hz **12 Hz**

1. two halves blink at different speeds



2. the visual cortex follows both

(here: focusing on the left)

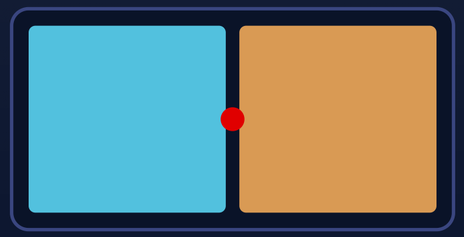


3. the one you focus on is stronger

HOW IT WORKS

How a test runs

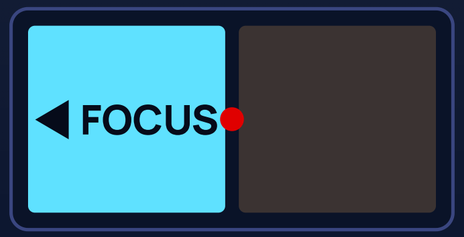
01



Check the signal

Both halves blink. We need to see **both** of them in the EEG. If we can't, the test stops and **no score** is shown.

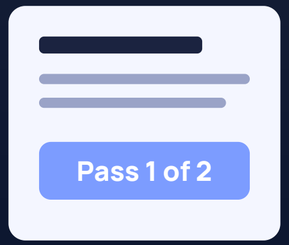
02



Give commands

"Focus on the left side." Eyes stay on the dot. **12 commands**, 6 left and 6 right, in random order. Only the EEG is used to score.

03



Score it

Pass mark is **10 of 12**. A referral needs **two passes in a row**, at least 12 hours apart.

SECOND TEST

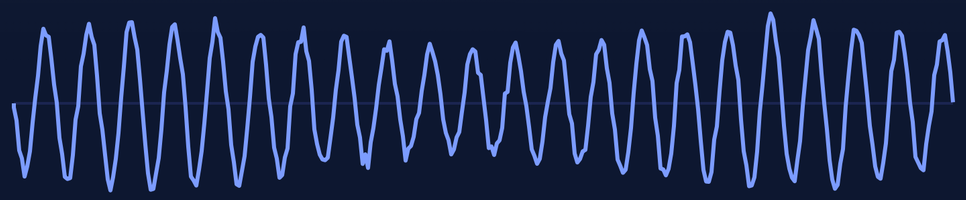
What if their eyes are closed?

A lot of patients don’t keep their eyes open. With eyes closed, the back of the brain shows strong alpha waves (8 to 13 Hz), and doing mental math usually makes them drop. So we built a second test that needs no screen at all.



"Just relax."

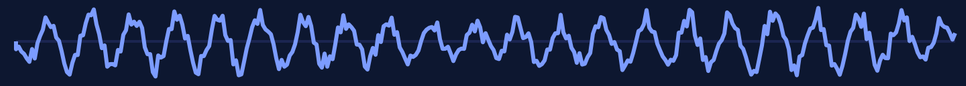
alpha stays high





"Count backwards from 600 by sevens."

alpha drops



12 commands (6 math, 6 relax) in random order. The 6 trials with the lowest alpha are called "math". Same rules as before: check the alpha rhythm first, pass at 10 of 12, two passes to refer. Note: in some people alpha goes up during math instead, so they'd just get "No conclusion".

RULES

Rules so one lucky run doesn't count for much

01 **No signal, no score.** If we can’t see the flicker or the sensors aren’t touching, it says "Not interpretable".

02 **Set before the test:** 12 trials, pass at 10, which sensors we use.

03 **No retrying until it passes.** One scored run per patient every 12 hours (the server checks this).

04 **Every run goes on the report,** including failed and stopped ones.

05 **It never says "conscious" or "unconscious".** Failing doesn’t mean nobody is there.

POSSIBLE RESULTS

Pass 1 of 2: repeat later.
Passed once. Needs another run.

Refer for full assessment.
Passed twice in a row.

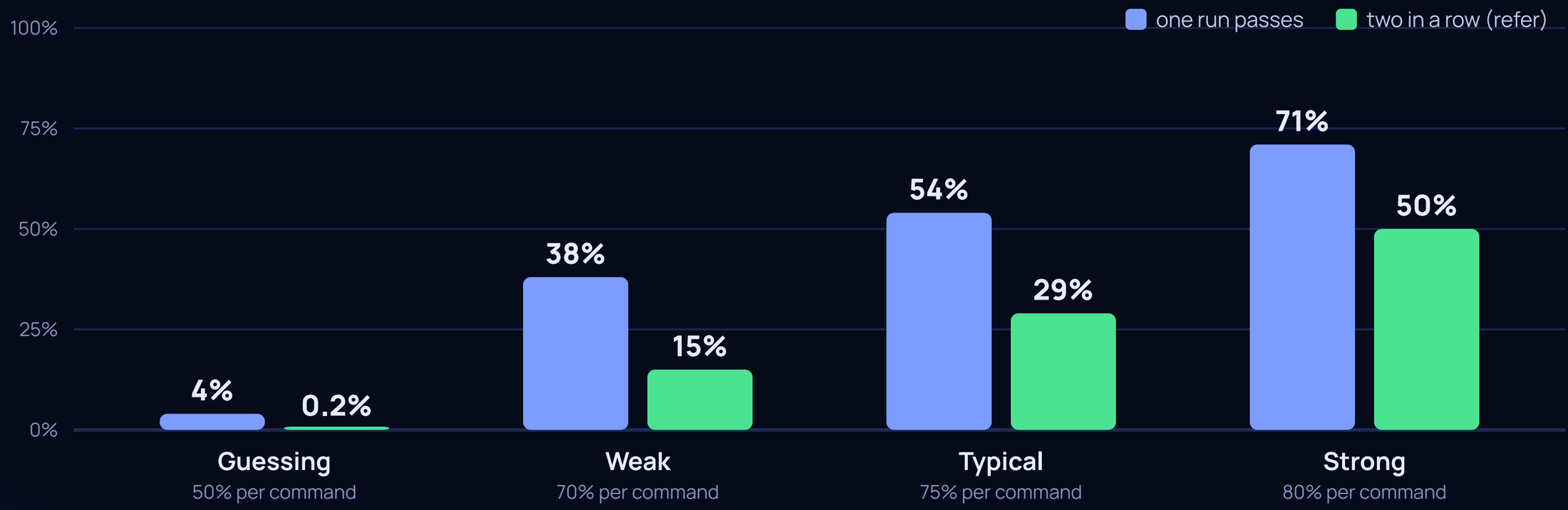
No conclusion.
Didn’t pass. Doesn’t rule anything out.

Not interpretable.
Signal not good enough. No score.

HOW STRICT IS IT?

Could someone pass by luck?

Pure guessing passes about 4% of the time, and 0.2% twice in a row. The downside is that real responders fail a lot too. We'd rather miss someone than wrongly pass them.



Guessing rate is an exact permutation test (6 left / 6 right). Other bars are from a simulation. Published attention-only accuracy is about 70 to 75% in healthy people.

REAL DATA

We also ran it on real EEG

The simulator only checks the code. So we ran our analysis on real recordings from g.tec's BR41N.IO SSVEP dataset (2 people, LEDs at 9, 10, 12 and 15 Hz), using just 4 of the 8 back-of-head channels, like our headset.

Person 1

39/40
picked the right flicker out of 4

20/20
9 vs 12 Hz with our left/right rule

2.6 vs 1.1
signal strength, flicker vs no flicker (step 1 needs 1.5+)

Person 2

26/40
picked the right flicker out of 4

16/20
9 vs 12 Hz with our left/right rule

1.65 vs 1.1
signal strength, flicker vs no flicker (step 1 needs 1.5+)

Note: in this dataset people looked straight at the LEDs, so it tests our signal processing on real EEG, not attention without looking. Person 2 would only just pass our step 1 check. Our own headset results are shown live in the demo.

PRIOR WORK

This builds on existing research

1996
Hillyard lab
Attention to one side boosts the flicker response. 8.6 and 12 Hz, left and right.

2014
Lesenfants et al.
Covert-attention SSVEP in locked-in syndrome. 24 healthy, 6 patients.

2014
Pan et al.
Visual awareness test in disorders of consciousness. 3 of 8 followed commands.

Today
g.tec mindBEAGLE
g.tec's clinical system. Uses sound, touch and imagined movement, 16 channels.

Nov 2026
StillHere (us)
Same idea on a 4-sensor dry headset and a laptop, plus the rules on slide 6.

Morgan, Hansen & Hillyard, PNAS 1996 · Lesenfants et al., J Neural Eng 2014 · Pan et al., J Neural Eng 2014 · g.tec mindBEAGLE product page

TECH

How we built it



The classifier sees EEG only. It never sees the command or the expected answer.

LIVE DEMO

Demo plan

We'll show the ways it should fail, not just the pass. For the first three, failing is the correct result.



Eyes closed

The flicker never reaches the visual cortex.

- Not interpretable



Ignore the voice

Count backward instead of following commands.

- No conclusion



Always left

Attend left whatever is asked. A stuck side can't pass.

- No conclusion



Follow each command

Eyes on the dot, a judge watching them. 10 or more of 12 to pass.

- Pass 1 of 2

LIMITATIONS

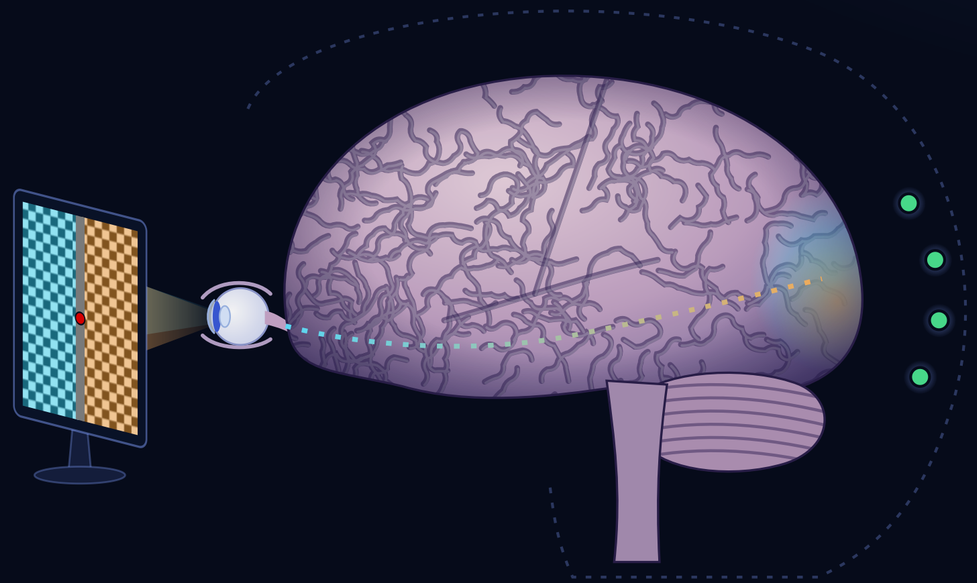
Limitations and next steps

Limitations

- We only tested it on ourselves
- Non-medical headset, 4 dry sensors
- Needs open eyes, working vision and language
- Real responders will often get "No conclusion"
- Flicker is not risk-free for photosensitive people

Next steps

- Test the **eyes-closed version** (built today) on more people
- Try it lying down (bed at 30°, pillow that keeps weight off the sensors)
- A proper study comparing it with fMRI or full EEG
- Clinical EEG supervision for any patient use



Thanks! Questions?

StillHere · github.com/shivanij1203/stillhere